

**KLAR**

Rx Only

COMPOSITIONS:
Clarithromycin Tablets 250 mg
 Each film - coated tablet contains
 Clarithromycin Ph.Eur. 250 mg.
Clarithromycin Tablets 500 mg
 Each film - coated tablet contains
 Clarithromycin Ph.Eur. 500 mg.

Adults

	Product	Component	Item Code	Date & Time
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 AUROBINDO Packaging Development		Customer / Country	Version No.	Reason Of Issue	24.04.2023 & 05.45 PM Reviewed / Approved by Sign / Date
		Malaysia_U3	14	Revision	
Team Leader	Kiran Kumar	Dimensions	No. of Colours : 01		
Initiator	Shirisha	210 x 480 mm			
Artist:	SCD	Pharmacode			
		16131			
Additional Information :					

Zidovudine

Simultaneous oral administration of clarithromycin tablets and zidovudine to HIV-infected adult patients may result in decreased steady-state zidovudine concentrations. Because clarithromycin appears to interfere with the absorption of simultaneously administered oral zidovudine, this interaction can be largely avoided by staggering the doses of clarithromycin and zidovudine to allow for a 4-hour interval between each medication. This interaction does not appear to occur in paediatric HIV-infected patients taking clarithromycin suspension with zidovudine or didoxynosine. This interaction is unlikely when clarithromycin is administered via intravenous infusion.

Phenytoin and Valproate

There have been spontaneous or published reports of interactions of CYP3A inhibitors, including clarithromycin with drugs not thought to be metabolised by CYP3A (e.g. phenytoin and valproate). Serum level determinations are recommended for these drugs when administered concomitantly with clarithromycin. Increased serum levels have been reported.

Bi-directional drug interactions:

Atazanavir

Both clarithromycin and atazanavir are substrates and inhibitors of CYP3A, and there is evidence of a bi-directional drug interaction. Co-administration of clarithromycin (500 mg twice daily) with atazanavir (400 mg once daily) resulted in a 2- fold increase in exposure to clarithromycin and a 70% decrease in exposure to 14-OH-clarithromycin, with a 28% increase in the AUC of atazanavir. Because of the large therapeutic window for clarithromycin, no dosage reduction should be necessary in patients with normal renal function. For patients with moderate renal function (creatinine clearance 30 to 60 mL/min), the dose of clarithromycin should be decreased by 50%. For patients with creatinine clearance <30 mL/min, the dose of clarithromycin should be decreased by 75% using an appropriate clarithromycin formulation. Doses of clarithromycin greater than 1000 mg per day should not be co-administered with protease inhibitors.

Calcium Channel Blockers

Caution is advised regarding the concomitant administration of clarithromycin and calcium channel blockers metabolized by CYP3A4 (e.g. verapamil, amiodipine, diltiazem) due to the risk of hypotension. Plasma concentrations of clarithromycin as well as calcium channel blockers may increase due to the interaction. Hypotension, bradyarrhythmias and lactic acidosis have been observed in patients taking clarithromycin and verapamil concomitantly.

Itraconazole

Both clarithromycin and itraconazole are substrates and inhibitors of CYP3A, leading to a bidirectional drug interaction. Clarithromycin may increase the plasma levels of itraconazole, while itraconazole may increase the plasma levels of clarithromycin. Patients taking itraconazole and clarithromycin concomitantly should be monitored closely for signs or symptoms of increased or prolonged pharmacologic effect.

Saquinavir

Both clarithromycin and saquinavir are substrates and inhibitors of CYP3A, and there is evidence of a bi-directional drug interaction. Concomitant administration of clarithromycin (500 mg twice daily) and saquinavir (soft gelatin capsules, 1200 mg three times daily) resulted in steady-state AUC and Cmax values of saquinavir which were177% and 187% higher than those seen with saquinavir alone. Clarithromycin AUC and Cmax values were approximately 40% higher than those seen with clarithromycin alone. No dose adjustment is required when the two drugs are co-administered for a limited time at the doses/formulations studied. Observations from drug interaction studies using the soft gelatin capsule formulation may not be representative of the effects seen using the saquinavir hard gelatin capsule. Observations from drug interaction studies performed with saquinavir alone may not be representative of the effects seen with saquinavir/ritonavir therapy. When saquinavir is co-administered with ritonavir, consideration should be given to the potential effects of ritonavir on clarithromycin.

Pregnancy and lactation

Pregnancy

The safety of clarithromycin for use during pregnancy has not been established. Based on variable results obtained from studies in mice, rats, rabbits and monkeys, the possibility of adverse effects on embryofoetal development cannot be excluded. Therefore, use during pregnancy is not advised without carefully weighing the benefits against risk.

Lactation

The safety of clarithromycin for use during breast feeding of infants has not been established. Clarithromycin is excreted into human breast milk.

Undesirable Effects

a. Summary of the safety profile

The most frequent and common adverse reactions related to clarithromycin therapy for both adult and pediatric populations are abdominal pain, diarrhea, nausea, vomiting and taste perversion. These adverse reactions are usually mild in intensity and are consistent with the known safety profile of macrolide antibiotics.

There was no significant difference in the incidence of these gastrointestinal adverse reactions between the patient population with or without preexisting mycobacterial infections.

System Organ Class	Common	Uncommon	Very Rare	Not Known
Infections and infestations:	Oral moniliasis	Candidiasis, vaginal infection		Pseudomembranous colitis, erysipelas
Blood and lymphatic system		Leukopenia, Neutropenia ¹ , Eosinophilia ¹		Agranulocytosis, Thrombocytopenia
Immune system disorders		Hypersensitivity		Anaphylactic reaction, angioedema
Metabolism and nutrition disorders		Anorexia, decreased appetite		
Psychiatric disorders	Insomnia	Anxiety	Nightmares	Psychotic disorder, confusional state, depersonalisation, depression, disorientation, hallucination, abnormal, dreams
Nervous system disorders	Dysgeusia, headache, taste perversion.	Dizziness, somnolence ² , tremor		Convulsion, ageusia, parosmia, anosmia , paraesthesia
Ear and labyrinth disorders		Vertigo, impaired hearing, tinnitus		deafness
Cardiac disorders				Torsade de Pointes, ventricular tachycardia, ventricular fibrillation
Vascular disorders				Hemorrhage
Gastrointestinal disorders	Diarrhea ³ , vomiting, dyspepsia, nausea, abdominal pain.	Gastritis, stomatitis, glossitis, abdominal distension ¹ , constipation, dry mouth, eructation, flatulence.		Pancreatitis, tongue discoloration, tooth discoloration.
Hepato-biliary disorders.	Liver function test abnormal.	Cholestasis ¹ , hepatitis ¹ , alanine aminotransferase increased, aspartate aminotransferase increased, gammaglutamyl transferase increased ¹ .		Hepatic failure, jaundice hepatocellular
Skin and subcutaneous tissue disorders:	Rash, hyperhidrosis, ,	pruritus, urticaria.		severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP).
Musculo skeletal, connective tissue and bone disorders:				Rhabdomyolysis ⁴ , Myopathy
Renal and urinary disorders:				Renal failure, interstitial nephritis.
General disorders and administration site conditions:		Malaise ¹ , asthenia ¹ , chest pain ¹ , chills ¹ , fatigue ¹ .		
Investigations	Elevated BUN (blood urea nitrogen)	Serum creatinine raised, blood alkaline phosphatase increased ¹ , blood lactate dehydrogenase increased ¹		International normalised ratio increased, prothrombin time prolonged, urine colour abnormal

¹ ADRs reported only for the Immediate-Release Tablets formulation

³ See section a)

^{2,4} See section b)

b. Description of selected adverse reactions

In some of the reports of rhabdomyolysis, clarithromycin was administered concomitantly with statins, fibrates, colchicine or allopurinol.

There have been rare reports of clarithromycin ER tablets in the stool, many of which have occurred in patients with anatomic (including ileostomy or colostomy) or functional gastrointestinal disorders with shortened GI transit times. In several reports, tablet residues have occurred in the context of diarrhea. It is recommended that patients who experience tablet residue in the stool and no improvement in their condition should be switched to a different clarithromycin formulation (e.g. suspension) or another antibiotic.

Special population: Adverse Reactions in Immunocompromised Patients

c. Paediatric populations

children under 12 years of age should use clarithromycin paediatric suspension. There are insufficient data to recommend a dosage regimen for use of the clarithromycin IV formulation in patients less than 18 years of age.

Frequency, type and severity of adverse reactions in children are expected to be the same as in adults.

d. Other special populations

Immunocompromised patients

In AIDS and other immunocompromised patients treated with the higher doses of clarithromycin over long periods of time for mycobacterial infections, it was often difficult to distinguish adverse events possibly associated with clarithromycin administration from underlying signs of Human Immunodeficiency Virus (HIV) disease or intercurrent illness.

In adult patients, the most frequently reported adverse reactions by patients treated with total daily doses of 1,000 mg and 2,000 mg of clarithromycin were: nausea, vomiting, taste perversion, abdominal pain, diarrhea, rash, flatulence, headache, constipation, hearing disturbance, Serum Glutamic Oxaloacetic Transaminase (SGOT) and Serum Glutamic Pyruvate Transaminase (SGPT) elevations. Additional low-frequency events included dyspnoea, insomnia and dry mouth.

The incidences were comparable for patients treated with 1,000 mg and 2,000 mg, but were generally about 3 to 4 times as frequent for those patients who received total daily doses of 4,000 mg of clarithromycin.

In these immunocompromised patients, evaluations of laboratory values were made by analysing those values outside the seriously abnormal level (i.e. the extreme high or low limit) for the specified test. On the basis of these criteria, about 2% to 3% of those patients who received 1,000 mg or 2,000 mg of clarithromycin daily had seriously abnormal elevated levels of SGOT and SGPT, and abnormally low white blood cell and platelet counts. A lower percentage of patients in these two dosage groups also had elevated Blood Urea Nitrogen levels. Slightly higher incidences of abnormal values were noted for patients who received 4,000 mg daily for all parameters except White Blood Cell.

Overdosage

Reports indicate that the ingestion of large amounts of clarithromycin can be expected to produce gastro-intestinal symptoms. One patient who had a history of bipolar disorder ingested 8 grams of clarithromycin and showed altered mental status, paranoid behaviour, hypokalemia and hypoxemia. Adverse reactions accompanying overdosage should be treated by gastric lavage and supportive measures. As with other macrolides, clarithromycin serum levels are not expected to be appreciably affected by haemodialysis or peritoneal dialysis.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Clarithromycin is a semi-synthetic derivative of erythromycin A.

Clarithromycin exerts its antibacterial action by binding to the 50s ribosomal sub-unit of susceptible bacteria and suppresses protein synthesis. It is highly potent against a wide variety of aerobic and anaerobic gram-positive and gram-negative organisms. The minimum inhibitory concentrations (MICs) of clarithromycin are generally two-fold lower than the MICs of erythromycin.

The 14-hydroxy metabolite of clarithromycin also has antimicrobial activity. The MICs of this metabolite are equal or two fold higher than the MICs of the parent compound, except for H influenzae where the 14-hydroxy metabolite is two-fold more active than the parent compound.

Susceptibility

The prevalence of resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. This information provides only an appropriate guidance on the probability of an organism being susceptible to clarithromycin.

Susceptible:

Commonly susceptible species
Aerobic Gram-negative microorganisms <i>Moraxella catarrhalis</i>
Anaerobic microorganisms <i>Peptococcus species</i> <i>Peptostreptococcus species</i> <i>Propionibacterium acnes</i> <i>Clostridium perfringens</i>
Other microorganisms <i>Chlamydia pneumoniae</i> <i>Legionella pneumophila</i> <i>Mycoplasma pneumoniae</i>

Species for which acquired resistance may be a problem
Aerobic Gram-positive microorganisms <i>Staphylococcus aureus</i> (methicillin-susceptible) <i>Staphylococcus aureus</i> (methicillin-resistant)+ <i>Streptococcus agalactiae</i> <i>Streptococcus pneumoniae</i> <i>Streptococcus pyogenes</i>
Aerobic Gram-negative microorganisms <i>Haemophilus influenzae</i> S

+ Resistance to macrolides among MRSA is commonly more than 50% in the EU and affects nearly all strains in some areas.

\$ The species may be regarded as being of intermediate susceptibility to clarithromycin and may also acquire high-level resistance to macrolides.

Resistance

Resistance mechanisms against macrolide antibiotics include alteration of the target site of the antibiotic or are based on modification and/or the active efflux of the antibiotic. Resistance development can be mediated via chromosomes or plasmids, be induced or exists constitutively. Macrolide-resistant bacteria generates enzymes which lead to methylation of residual adenine at ribosomal RNA and consequently to inhibition of the antibiotic binding to the ribosome. Macrolide/resistant organisms are generally cross/resistant to lincosamides and streptogramin B based on methylation of the ribosomal binding site. Clarithromycin ranks among the strong inducers of this enzyme as well. Furthermore, macrolides have a bacteriostatic action by inhibiting the peptidyl transferase of ribosomes. A complete cross-resistance exists among clarithromycin, erythromycin and azithromycin. Methicillin/ resistant staphylococci and penicillin-resistant *Streptococcus pneumoniae* are resistant to macrolides such as clarithromycin.

Pharmacokinetic properties

Clarithromycin is rapidly and well absorbed from the gastro-intestinal tract after oral administration of Clarithromycin tablets. The microbiologically active metabolite 14-hydroxycarithromycin is formed by first pass metabolism. Clarithromycin Tablets may be given without regard to meals as food does not affect the extent of bioavailability of Clarithromycin tablets. Food does slightly delay the onset of absorption of clarithromycin and formation of the 14-hydroxymetabolite. The pharmacokinetics of clarithromycin are non- linear; however, steadystate is attained within 2 days of dosing. At 250 mg b.i.d. 15-20% of unchanged drug is excreted in the urine. With 500 mg b.i.d. daily dosing urinary excretion is greater (approximately 36%). The 14-hydroxycarithromycin is the major urinary metabolite and accounts for 10-15% of the dose. Most of the remainder of the dose is eliminated in the faeces, primarily via the bile. 5-10% of the parent drug is recovered from the faeces.

When clarithromycin 500 mg is given three times daily, the clarithromycin plasma concentrations are increased with respect to the 500 mg twice daily dosage.

Clarithromycin provides tissue concentrations that are several times higher than the circulating drug levels.

Increased levels have been found in both tonsillar and lung tissue. Clarithromycin is 80% bound to plasma proteins at therapeutic levels.

Clarithromycin also penetrates the gastric mucus. Levels of clarithromycin in gastric mucus and gastric tissue are higher when clarithromycin is co-administered with omeprazole than when clarithromycin is administered alone.

PHARMACEUTICAL PARTICULARS

List of excipients

Cellulose Microcrystalline, Croscarmellose sodium, Povidone, Silica,colloidal anhydrous, Magnesium stearate & Opadry yellow.

Incompatibilities

None known.

Shelf life

24 Months

Special precautions for storage

Store in a dry place below 30°C.

Nature and contents of container

Clarithromycin Tablets available in Clear PVC/PVdC - Aluminium blister pack.

Blister pack:
10 Tablets (10's Blister X 1)

Manufactured By:



AUROBINDO
AUROBINDO PHARMA LIMITED,
Unit III Survey No 313 314, Bachupally,
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Product Registration Holder in Malaysia
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DATE OF REVISION OF THIS LEAFLET
21.04.2023